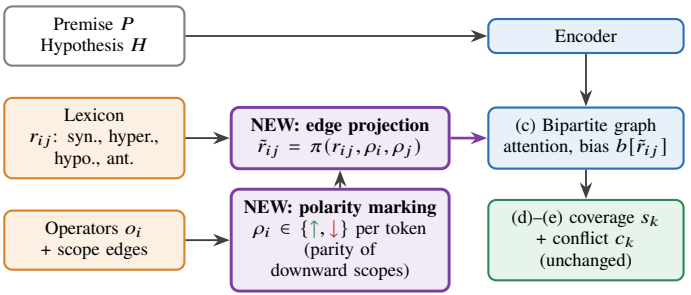
(a) Where v2 acts in COVER-PL



No new parameters: only the relation id fed to the graph changes.
Ablation = switch the projection off ($\tilde{r}_{ij} = r_{ij}$, i.e. v1).

(b) Polarity marking ρ

Upward: specific ⇒ general

A ↑ dog ↑ barks ↑

Downward: general ⇒ specific

No animal ↓ barks ↓

Every animal ↓ barks ↑

He did **not** eat ↓ fruit ↓

It is **not** true that he did **not** eat ↑

downward operators: *no, not, few, at most, without, never*; restrictor of *every/all*. Two downward scopes cancel (↑ again).

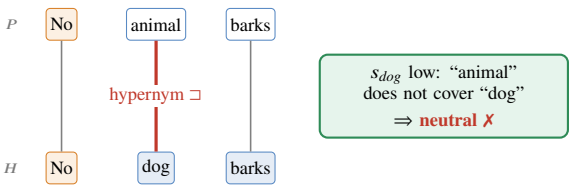
(c) Edge projection π (when $\rho_i = \rho_j$)

lexical edge $p_i \to h_j$	context ↑	context ↓
synonym (≡)	≡	≡
hyponym, *dog→animal* (⊏)	⊏ supports	⊐ does not
hypernym, *animal→dog* (⊐)	⊐ does not	⊏ supports
antonym (|)	|	| (conflict head)

⊏ = premise word is more specific (forward entailment). If $\rho_i \neq \rho_j$, the edge is left unchanged and the polarity mismatch is passed as an operator feature.

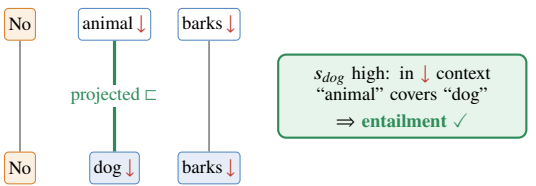
(d) Example: *P = No animal barks* *H = No dog barks* gold: entailment

v1 (current)



s_{dog} low: "animal" does not cover "dog"
⇒ **neutral ✗**

v2 (polarity projection)



s_{dog} high: in ↓ context "animal" covers "dog"
⇒ **entailment ✓**

Hypothesis to test

- InferBR / SICK: little change (downward contexts are rare)
- AX-b: downward monotone (v1 MCC −0.34), restrictivity, negation
- MED (monotonicity, unseen): main target
- ablation: v2 vs. v1, 10 paired seeds